
Beginner's Essential

Javascript Cheat Sheet

The language of the web.

Table of Contents

Javascript Basics	2
Variables	2
Arrays	3
Operators	4
Functions	5
Loops	7
If - Else Statements	7
Strings	7
Regular Expressions	9
Numbers and Math	10
Dealing with Dates	12
DOM Node	14
Working with the Browser	18
Events	21
Errors	27

Javascript Basics

Including JavaScript in an HTML Page

```
<script type="text/javascript">  
  //JS code goes here  
</script>
```

Call an External JavaScript File

```
<script src="myscript.js"></script><code></code>
```

Including Comments

```
//
```

Single line comments

```
/* comment here */
```

Multi-line comments

Variables

var, const, let

var

The most common variable. Can be reassigned but only accessed within a function. Variables defined with var move to the top when code is executed.

const

Cannot be reassigned and not accessible before they appear within the code.

let

Similar to const, however, let variable can be reassigned but not re-declared.

Data Types

```
var age = 23
```

Numbers

```
var x
```

Variables

```
var a = "init"
```

Text (strings)

```
var b = 1 + 2 + 3
```

Operations

```
var c = true
```

True or false statements

```
const PI = 3.14
```

Constant numbers

```
var name = {firstName:"John", lastName:"Doe"}
```

Objects

Objects

```
var person = {  
  firstName:"John",  
  lastName:"Doe",  
  age:20,  
  nationality:"German"  
};
```

Arrays

```
var fruit = ["Banana", "Apple", "Pear"];
```

Array Methods

```
concat()
```

Join several arrays into one

```
indexOf()
```

Returns the first position at which a given element appears in an array

```
join()
```

Combine elements of an array into a single string and return the string

```
lastIndexOf()
```

Gives the last position at which a given element appears in an array

pop()

Removes the last element of an array

push()

Add a new element at the end

reverse()

Reverse the order of the elements in an array

shift()

Remove the first element of an array

slice()

Pulls a copy of a portion of an array into a new array of 4 24

sort()

Sorts elements alphabetically

splice()

Adds elements in a specified way and position

toString()

Converts elements to strings

unshift()

Adds a new element to the beginning

valueOf()

Returns the primitive value of the specified object

Operators

Basic Operators

+	Addition
-	Subtraction
*	Multiplication
/	Division
(...)	Grouping operator
%	Modulus (remainder)
++	Increment numbers
--	Decrement numbers

Comparison Operators

```
==    Equal to
===   Equal value and equal type
!=    Not equal
!==   Not equal value or not equal type
>     Greater than
<     Less than
>=    Greater than or equal to
<=    Less than or equal to
?     Ternary operator
```

Logical Operators

```
&&    Logical and
||     Logical or
!      Logical not
```

Bitwise Operators

```
&      AND statement
|      OR statement
~      NOT
^      XOR
<<     Left shift
>>     Right shift
>>>    Zero fill right shift
```

Functions

```
function name(parameter1, parameter2, parameter3) {
    // what the function does
}
```

Outputting Data

alert()

Output data in an alert box in the browser window

confirm()

Opens up a yes/no dialog and returns true/false depending on user click

console.log()

Writes information to the browser console, good for debugging purposes

`document.write()`

Write directly to the HTML document

`prompt()`

Creates an dialogue for user input

Global Functions

`decodeURI()`

Decodes a Uniform Resource Identifier (URI) created by `encodeURIComponent` or similar

`decodeURIComponent()`

Decodes a URI component

`encodeURIComponent()`

Encodes a URI into UTF-8

`encodeURIComponent()`

Same but for URI components

`eval()`

Evaluates JavaScript code represented as a string

`isFinite()`

Determines whether a passed value is a finite number

`isNaN()`

Determines whether a value is NaN or not

`Number()`

Returns a number converted from its argument

`parseFloat()`

Parses an argument and returns a floating point number

`parseInt()`

Parses its argument and returns an integer

Loops

```
for (before loop; condition for loop; execute after loop) {  
    // what to do during the loop  
}  
for
```

The most common way to create a loop in Javascript

while

Sets up conditions under which a loop executes

do while

Similar to the while loop, however, it executes at least once and performs a check at the end to see if the condition is met to execute again

break

Used to stop and exit the cycle at certain conditions

continue

Skip parts of the cycle if certain conditions are met of 724

If - Else Statements

```
if (condition) {  
    // what to do if condition is met  
} else {  
    // what to do if condition is not met  
}
```

Strings

```
var person = "John Doe";
```

Escape Characters

```
\'    - Single quote  
\"    - Double quote  
\\    - Backslash  
\\b   - Backspace  
\\f   - Form feed  
\\n   - New line  
\\r   - Carriage return  
\\t   - Horizontal tabulator
```


`\v` – Vertical tabulator

String Methods

`charAt()`

Returns a character at a specified position inside a string

`charCodeAt()`

Gives you the unicode of character at that position

`concat()`

Concatenates (joins) two or more strings into one

`fromCharCode()`

Returns a string created from the specified sequence of UTF-16 code units

`indexOf()`

Provides the position of the first occurrence of a specified text within a string

`lastIndexOf()`

Same as `indexOf()` but with the last occurrence, searching backwards

`match()`

Retrieves the matches of a string against a search pattern

`replace()`

Find and replace specific text in a string

`search()`

Executes a search for a matching text and returns its position

`slice()`

Extracts a section of a string and returns it as a new string

`split()`

Splits a string object into an array of strings at a specified position

`substr()`

Similar to `slice()` but extracts a substring depended on a specified number of characters

`substring()`

Also similar to `slice()` but can't accept negative indices

`toLowerCase()`

Convert strings to lowercase

`toLowerCase()`

Convert strings to uppercase

`valueOf()`

Returns the primitive value (that has no properties or methods) of a string object

Regular Expressions

Pattern Modifiers

`e` – Evaluate replacement
`i` – Perform case-insensitive matching
`g` – Perform global matching
`m` – Perform multiple line matching
`s` – Treat strings as single line
`x` – Allow comments and whitespace in pattern
`U` – Non Greedy pattern

Brackets

`[abc]` Find any of the characters between the brackets
`[^abc]` Find any character not in the brackets
`[0-9]` Used to find any digit from 0 to 9
`[A-z]` Find any character from uppercase A to lowercase z
`(a|b|c)` Find any of the alternatives separated with |

Metacharacters

`.` – Find a single character, except newline or line terminator
`\w` – Word character
`\W` – Non-word character
`\d` – A digit
`\D` – A non-digit character
`\s` – Whitespace character
`\S` – Non-whitespace character
`\b` – Find a match at the beginning/end of a word
`\B` – A match not at the beginning/end of a word
`\0` – NUL character
`\n` – A new line character
`\f` – Form feed character
`\r` – Carriage return character
`\t` – Tab character
`\v` – Vertical tab character

`\xxx` – The character specified by an octal number `xxx`
`\xdd` – Character specified by a hexadecimal number `dd`
`\uxxxx` – The Unicode character specified by a hexadecimal number `xxxx`

Quantifiers

`n+` – Matches any string that contains at least one `n`
`n*` – Any string that contains zero or more occurrences of `n`
`n?` – A string that contains zero or one occurrences of `n`
`n{X}` – String that contains a sequence of `X` `n`'s
`n{X,Y}` – Strings that contains a sequence of `X` to `Y` `n`'s
`n{X,}` – Matches any string that contains a sequence of at least `X` `n`'s
`n$` – Any string with `n` at the end of it
`^n` – String with `n` at the beginning of it
`?=n` – Any string that is followed by a specific string `n`
`?!n` – String that is not followed by a specific string `n`

Numbers and Math

Number Properties

`MAX_VALUE`

The maximum numeric value representable in JavaScript

`MIN_VALUE`

Smallest positive numeric value representable in JavaScript

`NaN`

The “Not-a-Number” value

`NEGATIVE_INFINITY`

The negative Infinity value

`POSITIVE_INFINITY`

Positive Infinity value

Number Methods

`toExponential()`

Returns a string with a rounded number written as exponential notation

`toFixed()`

Returns the string of a number with a specified number of decimals

toFixed()

String of a number written with a specified length

toString()

Returns a number as a string

valueOf()

Returns a number as a number

Math Properties

E	Euler's number
LN2	The natural logarithm of 2
LN10	Natural logarithm of 10
LOG2E	Base 2 logarithm of E
LOG10E	Base 10 logarithm of E
PI	The number PI
SQRT1_2	Square root of 1/2
SQRT2	The square root of 2

Math Methods**abs(x)**

Returns the absolute (positive) value of x

acos(x)

The arccosine of x, in radians

asin(x)

Arcsine of x, in radians

atan(x)

The arctangent of x as a numeric value

atan2(y,x)

Arctangent of the quotient of its arguments

ceil(x)

Value of x rounded up to its nearest integer

cos(x)

The cosine of x (x is in radians)

exp(x)

Value of Ex

floor(x)

The value of x rounded down to its nearest integer

log(x)

The natural logarithm (base E) of x

max(x,y,z,...,n)

Returns the number with the highest value

min(x,y,z,...,n)

Same for the number with the lowest value

pow(x,y)

X to the power of y

random()

Returns a random number between 0 and 1

round(x)

The value of x rounded to its nearest integer

sin(x)

The sine of x (x is in radians)

sqrt(x)

Square root of x

tan(x)

The tangent of an angle

Dealing with Dates

Setting Dates

Date()

Creates a new date object with the current date and time

Date(2017, 5, 21, 3, 23, 10, 0)

Create a custom date object. The numbers represent year, month, day, hour, minutes, seconds, milliseconds. You can omit anything you want except for year and month.

Date("2017-06-23")

Date declaration as a string

Pulling Date and Time Values

getDate()

Get the day of the month as a number (1-31)

getDay()

The weekday as a number (0-6)

getFullYear()

Year as a four digit number (yyyy)

getHours()

Get the hour (0-23)

getMilliseconds()

The millisecond (0-999)

getMinutes()

Get the minute (0-59)

getMonth()

Month as a number (0-11)

getSeconds()

Get the second (0-59)

getTime()

Get the milliseconds since January 1, 1970

getUTCDate()

The day (date) of the month in the specified date according to universal time (also available for day, month, fullyear, hours, minutes etc.)

parse

Parses a string representation of a date, and returns the number of milliseconds since January 1, 1970

Set Part of a Date

`setDate()`

Set the day as a number (1-31)

`setFullYear()`

Sets the year (optionally month and day)

`setHours()`

Set the hour (0-23)

`setMilliseconds()`

Set milliseconds (0-999)

`setMinutes()`

Sets the minutes (0-59)

`setMonth()`

Set the month (0-11)

`setSeconds()`

Sets the seconds (0-59)

`setTime()`

Set the time (milliseconds since January 1, 1970)

`setUTCDate()`

Sets the day of the month for a specified date according to universal time (also available for day, month, fullyear, hours, minutes etc.)

DOM Node

Node Properties

`attributes`

Returns a live collection of all attributes registered to and element

`baseURI`

Provides the absolute base URL of an HTML element

`childNodes`

Gives a collection of an element's child nodes

firstChild

Returns the first child node of an element

lastChild

The last child node of an element

nextSibling

Gives you the next node at the same node tree level

nodeName

Returns the name of a node

nodeType

Returns the type of a node

nodeValue

Sets or returns the value of a node

ownerDocument

The top-level document object for this node

parentNode

Returns the parent node of an element

previousSibling

Returns the node immediately preceding the current one

textContent

Sets or returns the textual content of a node and its descendants

Node Methods**appendChild()**

Adds a new child node to an element as the last child node

cloneNode()

Clones an HTML element

compareDocumentPosition()

Compares the document position of two elements

getFeature()

Returns an object which implements the APIs of a specified feature

hasAttributes ()

Returns true if an element has any attributes, otherwise false

hasChildNodes ()

Returns true if an element has any child nodes, otherwise false

insertBefore ()

Inserts a new child node before a specified, existing child node

isDefaultNamespace ()

Returns true if a specified namespaceURI is the default, otherwise false

isEqualNode ()

Checks if two elements are equal

isSameNode ()

Checks if two elements are the same node

isSupported ()

Returns true if a specified feature is supported on the element

lookupNamespaceURI ()

Returns the namespaceURI associated with a given node

lookupPrefix ()

Returns a DOMString containing the prefix for a given namespaceURI, if present

normalize ()

Joins adjacent text nodes and removes empty text nodes in an element

removeChild ()

Removes a child node from an element

replaceChild ()

Replaces a child node in an element

Element Methods**getAttribute ()**

Returns the specified attribute value of an element node

getAttributeNS ()

Returns string value of the attribute with the specified namespace and name

getAttributeNode ()

Gets the specified attribute node

getAttributeNodeNS ()

Returns the attribute node for the attribute with the given namespace and name

getElementsByTagName ()

Provides a collection of all child elements with the specified tag name

getElementsByTagNameNS ()

Returns a live HTMLCollection of elements with a certain tag name belonging to the given namespace

hasAttribute ()

Returns true if an element has any attributes, otherwise false

hasAttributeNS ()

Provides a true/false value indicating whether the current element in a given namespace has the specified attribute

removeAttribute ()

Removes a specified attribute from an element

removeAttributeNS ()

Removes the specified attribute from an element within a certain namespace

removeAttributeNode ()

Takes away a specified attribute node and returns the removed node

setAttribute ()

Sets or changes the specified attribute to a specified value

setAttributeNS ()

Adds a new attribute or changes the value of an attribute with the given namespace and name

setAttributeNode ()

Sets or changes the specified attribute node

setAttributeNodeNS ()

Adds a new namespaced attribute node to an element

Working with the Browser

Window Properties

closed

Checks whether a window has been closed or not and returns true or false

defaultStatus

Sets or returns the default text in the statusbar of a window

document

Returns the document object for the window

frames

Returns all <iframe> elements in the current window

history

Provides the History object for the window

innerHeight

The inner height of a window's content area

innerWidth

The inner width of the content area

length

Find out the number of <iframe> elements in the window

location

Returns the location object for the window

name

Sets or returns the name of a window

navigator

Returns the Navigator object for the window

opener

Returns a reference to the window that created the window

outerHeight

The outer height of a window, including toolbars/ scrollbars

outerWidth

The outer width of a window, including toolbars/ scrollbars

pageXOffset

Number of pixels the current document has been scrolled horizontally

pageYOffset

Number of pixels the document has been scrolled vertically

parent

The parent window of the current window

screen

Returns the Screen object for the window

screenLeft

The horizontal coordinate of the window (relative to screen)

screenTop

The vertical coordinate of the window

screenX

Same as screenLeft but needed for some browsers

screenY

Same as screenTop but needed for some browsers

self

Returns the current window

status

Sets or returns the text in the statusbar of a window

top

Returns the topmost browser window

Window Methods**alert()**

Displays an alert box with a message and an OK button

blur()

Removes focus from the current window

clearInterval()

Clears a timer set with setInterval()

clearTimeout()

Clears a timer set with setTimeout()

close()

Closes the current window

confirm()

Displays a dialogue box with a message and an OK and Cancel button

focus()

Sets focus to the current window

moveBy()

Moves a window relative to its current position

moveTo()

Moves a window to a specified position

open()

Opens a new browser window

print()

Prints the content of the current window

prompt()

Displays a dialogue box that prompts the visitor for input

resizeBy()

Resizes the window by the specified number of pixels

resizeTo()

Resizes the window to a specified width and height

scrollBy()

Scrolls the document by a specified number of pixels

scrollTo()

Scrolls the document to specific coordinates

setInterval()

Calls a function or evaluates an expression at specified intervals

setTimeout()

Calls a function or evaluates an expression after a specified interval

stop()

Stops the window from loading

Screen Properties**availHeight**

Returns the height of the screen (excluding the Windows Taskbar)

availWidth

Returns the width of the screen (excluding the Windows Taskbar)

colorDepth

Returns the bit depth of the color palette for displaying images

height

The total height of the screen

pixelDepth

The color resolution of the screen in bits per pixel

width

The total width of the screen

Events

Mouse**onclick**

The event occurs when the user clicks on an element

oncontextmenu

User right-clicks on an element to open a context menu

ondblclick

The user double-clicks on an element

onmousedown

User presses a mouse button over an element

onmouseenter

The pointer moves onto an element

onmouseleave

Pointer moves out of an element

onmousemove

The pointer is moving while it is over an element

onmouseover

When the pointer is moved onto an element or one of its children

onmouseout

User moves the mouse pointer out of an element or one of its children

onmouseup

The user releases a mouse button while over an element

Keyboard**onkeydown**

When the user is pressing a key down

onkeypress

The moment the user starts pressing a key

onkeyup

The user releases a key

Frame**onabort**

The loading of a media is aborted

onbeforeunload

Event occurs before the document is about to be unloaded

onerror

An error occurs while loading an external file

onhashchange

There have been changes to the anchor part of a URL

onload

When an object has loaded

onpagehide

The user navigates away from a webpage

onpageshow

When the user navigates to a webpage

onresize

The document view is resized

onscroll

An element's scrollbar is being scrolled

onunload

Event occurs when a page has unloaded

Form**onblur**

When an element loses focus

onchange

The content of a form element changes (for <input>, <select>and <textarea>)

onfocus

An element gets focus

onfocusin

When an element is about to get focus

onfocusout

The element is about to lose focus

oninput

User input on an element

oninvalid

An element is invalid

onreset

A form is reset

onsearch

The user writes something in a search field (for `<input="search">`)

onselect

The user selects some text (for `<input>` and `<textarea>`)

onsubmit

A form is submitted

Drag**ondrag**

An element is dragged

ondragend

The user has finished dragging the element

ondragenter

The dragged element enters a drop target

ondragleave

A dragged element leaves the drop target

ondragover

The dragged element is on top of the drop target

ondragstart

User starts to drag an element

ondrop

Dragged element is dropped on the drop target

Clipboard**oncopy**

User copies the content of an element

oncut

The user cuts an element's content

onpaste

A user pastes content in an element

Media**onabort**

Media loading is aborted

oncanplay

The browser can start playing media (e.g. a file has buffered enough)

oncanplaythrough

When browser can play through media without stopping

ondurationchange

The duration of the media changes

onended

The media has reached its end

onerror

Happens when an error occurs while loading an external file

onloadeddata

Media data is loaded

onloadedmetadata

Meta Metadata (like dimensions and duration) are loaded

onloadstart

Browser starts looking for specified media

onpause

Media is paused either by the user or automatically

onplay

The media has been started or is no longer paused

onplaying

Media is playing after having been paused or stopped for buffering

onprogress

Browser is in the process of downloading the media

onratechange

The playing speed of the media changes

onseeked

User is finished moving/skipping to a new position in the media

onseeking

The user starts moving/skipping

onstalled

The browser is trying to load the media but it is not available

onsuspend

Browser is intentionally not loading media

ontimeupdate

The playing position has changed (e.g. because of fast forward)

onvolumechange

Media volume has changed (including mute)

onwaiting

Media paused but expected to resume (for example, buffering)

Animation**animationend**

A CSS animation is complete

animationiteration

CSS animation is repeated

animationstart

CSS animation has started

Other**transitionend**

Fired when a CSS transition has completed

onmessage

A message is received through the event source

onoffline

Browser starts to work offline

ononline

The browser starts to work online

onpopstate

When the window's history changes

onshow

A <menu> element is shown as a context menu

onstorage

A Web Storage area is updated

ontoggle

The user opens or closes the <details> element

onwheel

Mouse wheel rolls up or down over an element

ontouchcancel

Screen touch is interrupted

ontouchend

User finger is removed from a touch screen

ontouchmove

A finger is dragged across the screen

ontouchstart

Finger is placed on touch screen

Errors

try

Lets you define a block of code to test for errors

catch

Set up a block of code to execute in case of an error

throw

Create custom error messages instead of the standard JavaScript errors

finally

Lets you execute code, after try and catch, regardless of the result

Error Name Values**name**

Sets or returns the error name

message

Sets or returns an error message in string from

EvalError

An error has occurred in the eval() function

RangeError

A number is “out of range”

ReferenceError

An illegal reference has occurred

SyntaxError

A syntax error has occurred

TypeError

A type error has occurred

URIError

An encodeURI() error has occurred